TrackShift Energy Intelligence

Product Requirements Document v2.0 — Theme: The Power Play (Race-Long Energy Strategy)

TrackShift 2026 Team Submission

August 2026 — Revision 2.0 (incorporates technical review pass)

Contents

Executive Summary

TrackShift Energy Intelligence is a race-long, adaptive energy-deployment platform built around a reduced-order, transparently-parameterized F1 vehicle/battery simulator, an uncertainty-aware receding-horizon optimizer, and a real-time strategy dashboard that explains — not just displays — every decision.

Novelty statement. TrackShift Energy Intelligence is not "MPC for battery management." It combines (1) race-scale energy forecasting under explicit uncertainty, (2) regulation-aware constrained optimization against the actual, cited 2026 FIA power-unit rules, (3) a dynamic strategic-reserve formulation that widens automatically under forecast uncertainty rather than using a fixed safety margin, (4) structural (not scripted) real-time replanning, and (5) a domain-agnostic optimization core that is demonstrated — not merely claimed — to run a second, real-world configuration (EV delivery-fleet battery budgeting) with no change to the optimizer itself.

What this system is. A research prototype demonstrating the engineering principle of *dynamic constrained resource allocation of a depleting, partially-rechargeable energy store, under uncertainty, against a hard deadline* — instantiated first against a Haas-inspired 2026 F1 reference vehicle, and second against an EV delivery-fleet configuration, both driven by the same optimization core (Section 29).

What this system explicitly does not claim (Section 5): it is not a digital twin of the real Haas VF-26, does not use proprietary Haas data, does not claim FIA-certified simulation accuracy, and does not claim to be a production-ready fleet-optimization product. It claims to correctly implement the *publicly documented* 2026 FIA energy-system constraints and to demonstrate the transfer principle credibly within hackathon scope.

How this PRD is organized. Part I frames the problem and the constraint set (including a dedicated, cited FIA 2026 Energy-System Constraints section with a provenance table). Part II specifies the system itself, including an explicit mathematical formulation of the optimization problem — state, control, disturbance, constraints, objective — which previous drafts of this PRD left implicit. Part III specifies how the system is evaluated as an engineering artifact, including ablations, sensitivity analysis, and reproducibility requirements. Part IV specifies the real-world transfer as a second working configuration of the same core, not a slide. Part V specifies the dashboard and demo. Part VI specifies execution: strict P0/P1/P2 priorities, a minimum winning prototype, the phased roadmap, and the risk register.

Part I — The Problem

1. TrackShift Context and Judging Strategy

[FACT] Plaksha University's TrackShift program is run in partnership with the Mphasis Foundation, whose parent company Mphasis is the official Digital Partner of the Toyota Gazoo Racing (TGR) Haas F1 Team; the program is explicitly framed around transferring motorsport engineering principles — precision, data-led decision-making, and applied engineering — into open, industry-ready solutions for mobility, infrastructure, and manufacturing, with winning teams receiving an internship and a visit to the Haas F1 facility in the UK. The organizers' framing — *"Motorsport is not the destination. Motorsport is the proof of concept"* — is treated in this PRD as the binding design constraint, not a tagline.

[ASSUMPTION] The Theme 1 brief text quoted throughout this document ("Build an adaptive AI model that dynamically calculates optimal, multi-lap battery deployment strategies across changing race conditions") is taken as given from the internal challenge packet; we could not independently verify a public copy of the exact 2026 theme wording, so it is treated as authoritative for design purposes but should be re-confirmed against the official packet before submission.

1.1 Explicit mapping to TrackShift's three design principles

TrackShift principle	How this system satisfies it	PRD section
Constraint is the Design Brief	The optimizer's constraint set is the actual, cited, versioned 2026 FIA power-unit energy regulation — not an invented simplification — plus vehicle, safety, and reserve constraints	§2 (regulation), §12 (mathematical formulation)
Real-World Impact	The same optimization core is demonstrated running a second, working, real-world configuration (EV delivery-fleet energy budgeting), with a concrete India-specific scenario and a measured impact calculation, not a "this could be used for…" slide	§29–31
Motorsport as Inspiration	F1 energy strategy is generalized into a domain-agnostic resource-allocation primitive (state, control, disturbance, constraint, objective) that is provably reusable, per §29's shared-core architecture	§12, §29

1.2 Research questions vs. engineering requirements

[DESIGN] Kept explicitly separate so the project does not drift into an unfocused research exercise:

Research questions (what we want to learn/demonstrate): Does explicit short-horizon prediction improve energy strategy over a no-prediction physics-only optimizer? Does MPC materially outperform rule-based heuristics under the *actual* FIA 2026 constraint set? How robust is the strategy to previously unseen track geometry? How much does uncertainty-aware reserve management help versus a fixed-margin reserve? How close does the online (real-time, no future information) strategy get to an offline oracle that has full future knowledge?

Engineering requirements (what must simply work, independent of the research questions' answers): the simulator must run and produce physically consistent telemetry; the optimizer must always return a decision or an explicit infeasibility signal (Section 20), never silently fail; the dashboard must update in real time and never lag the control loop; hard constraints (SOC, regulatory power limits, reserve) must never be silently violated.

2. FIA 2026 Energy-System Constraints

This is the literal, load-bearing constraint set the optimizer is built against (Section 12). Every parameter below is version-tagged and sourced; **the previously-drafted PRD's flat "8 MJ regen" figure is corrected here and superseded by the versioned table in §2.5** — regeneration/deployment limits are not a single constant, they are venue- and session-dependent within a regulatory band, and the FIA revised the applicable articles as recently as August 5, 2026.

2.1 Regulation provenance and revision tracking

[DESIGN — required going forward] The system must record which regulation revision it was built against, and must **never hard-code** the numeric limits below as literals inside simulator or optimizer code. All values in this section are loaded from a single configuration object:

```
FIA_REGULATION_VERSION      = "2026 Technical Regulations, as amended 2026-08-05"
MAX_MGU_K_DEPLOY_POWER_KW   = 350        # baseline race deployment cap
MAX_MGU_K_REGEN_POWER_KW    = 350        # baseline race regen cap
MAX_ENERGY_RECOVERY_MJ_LAP  = {"default": 8.5, "min_event": 5.0, "max_event": 9.0}
ENERGY_STORE_CAP_MJ         = 4.0        # max usable battery energy at any time
DEPLOYMENT_CURVE            = [          # speed (km/h) -> max deploy power (kW)
  {"speed_kmh": 0,   "power_kw": 350},
  {"speed_kmh": 290, "power_kw": 350},
  {"speed_kmh": 355, "power_kw": 0}
]
LIMITED_POWER_MODE_CURVE    = [          # optional per-event "limited power" mode
  {"speed_kmh": 0,   "power_kw": 250},
  {"speed_kmh": 310, "power_kw": 250},
  {"speed_kmh": 340, "power_kw": 100},
  {"speed_kmh": 345, "power_kw": 0}
]
DEPLOYMENT_ZONING           = {"acceleration_zones_kw": 350, "elsewhere_kw": 250}
OVERRIDE_ASSIST_MJ          = 0.5        # following-car override, <1s gap
BOOST_MAX_POWER_KW          = 150        # incremental boost cap (race), per §2.4
```

This config is the single source of truth consumed by the vehicle/battery model (Sections 10–11) and the optimizer's constraint set (Section 12); nothing downstream re-derives or duplicates these numbers.

2.2 Regulation provenance table

Parameter	Value used in sim	Source	Regulation article / revision	Confidence
MGU-K max electrical deployment power	350 kW	Formula1.com, Honda Global, Red Bull Racing, ESPN, GPFans regulation coverage	2026 Technical Regulations (baseline, issued through mid-2025/2026 season)	**Official**
MGU-K max regeneration power	350 kW	Same as above	Same	**Official**

Parameter	Value used in sim	Source	Regulation article / revision	Confidence
Energy Store usable capacity	4 MJ	Same as above (ESPN, GPFans, Honda)	Same	**Official**
Baseline max energy recovery per lap (race)	8.5 MJ nominal, FIA-adjustable 5–9 MJ per venue	Raceteq, Motorsport Technology, Sports of the Day coverage of May–June 2026 WMSC refinements	2026 Technical Regs, mid-season refinement (~May 2026)	**Official**, venue-adjustable figure
“Limited power” mode curve (250 kW to 310 km/h, tapering to 0 by 345 km/h)	Modeled as an optional event-specific mode, **not** the default race curve	PaddockNews24, citing FIA Article C5.2.8 iii, published 2026-08-05	2026 Technical Regs, **August 5, 2026 amendment**	**Official (secondary source) —** applies only in designated sectors/sessions per Sporting Reg. Article B7.2, not a universal replacement of the 350 kW baseline
Sprint/Qualifying max recoverable energy reduced 5→4 MJ	Not applied to race-distance simulation (per source, this change does not affect races)	Same August 5, 2026 source	Article C5.2.10 ii	**Official (secondary source)**, session-scoped
Deployment taper to zero by ~355 km/h (baseline race curve)	Used as the default `DEPLOYMENT_CURVE`	ESPN 2026 regulation coverage	2026 Technical Regs baseline	**Official**
Override assist (following car, <1s gap)	≈0.5 MJ extra, up to 350 kW to ~337 km/h	Formula1.com explainer	2026 Technical Regs	**Official**
Boost button incremental cap (race)	+150 kW over current power level	Speedcafe.com, mid-season rules coverage	2026 mid-season refinement (~April 2026)	**Official (secondary source)**
MGU-K deployment zoning (350 kW in acceleration zones / 250 kW elsewhere)	Optional zoning mode, configurable	Speedcafe.com	2026 mid-season refinement	**Official (secondary source)**

Parameter	Value used in sim	Source	Regulation article / revision	Confidence
Minimum car weight	768–770 kg (sources differ by draft/date)	Red Bull Racing, GPFans, F1 Las Vegas GP, Formula1.com	2026 Technical Regs	**Official**, minor cross-source variance flagged
2027 power-split rule change (60/40, +20kW ICE, −50kW ERS, +25kW/lap harvest)	**Not used** — explicitly out of scope, this is a 2027 rule, not 2026	PlanetF1, June 2026	Ratified by WMSC for 2027 onward	**Official**, explicitly excluded as future/inapplicable
Vehicle mass, drag, tire/cornering, battery efficiency, thermal behavior	See §8 (Known/Assumed/Derived table)	—	—	**Engineering assumption**

2.3 What is genuinely "official 2026 regulation" vs. what this project assumes

[DESIGN] This distinction is the single most defensible thing in the whole PRD and should be repeated in the pitch: **every power/energy limit number is real, cited, current FIA regulation** (Section 2.1–2.2); **every vehicle-body and battery-chemistry number** (mass distribution, drag coefficient, cornering grip, round-trip efficiency, thermal derating) **is a labeled engineering assumption**, because those are proprietary to each team and were never public. The simulator's constraint boundaries are FIA-real; its vehicle-body physics are estimated and labeled as such.

2.4 Note on the "limited power" mode and why it is not the default

[DESIGN] The August 5, 2026 FIA update introduces a lower, speed-dependent power ceiling (`LIMITED_POWER_MODE_CURVE`) that applies only in designated sectors/sessions per sporting-regulation Article B7.2 — this reads as an FIA-administered energy-saving mode (e.g. for specific events or sectors), not a replacement of the season-baseline 350 kW deployment figure used throughout prior public regulation reporting. **[DESIGN]** the simulator implements both curves as selectable configs (`DEPLOYMENT_CURVE` as default, `LIMITED_POWER_MODE_CURVE` as an optional per-event override), and the scenario injector (Section 17) can toggle to the limited-power curve mid-race as one of its supported disturbance types — which conveniently doubles as a second, regulation-grounded "unexpected condition" for the adaptive-replanning demo, distinct from the safety-car/rain scenarios.

2.5 Regulation currency caveat

[DESIGN] FIA 2026 technical regulations have been amended multiple times within the 2026 season itself (mid-season refinements around April–June 2026, and again August 5, 2026). The team must re-check `FIA_REGULATION_VERSION` against the latest FIA/Formula1.com publication in the days immediately before the event and update the config file — not the code — if anything has changed again. This is precisely why these numbers are configuration, never hard-coded literals (Section 2.1).

3. Team Capability Mapping

Capability needed	Team member	Direct prior experience
Vehicle state, track geometry, planning/control loop, simulation & telemetry tooling	Member 1 (Formula Manipal Driverless)	CARLA, Gazebo, Foxglove, SLAM, YOLO, Delaunay triangulation, PID, Stanley controller, MPC, AV perception/localization/planning/control
Battery/energy modeling, SOC estimation, energy constraints, battery-model validation (§26)	Member 2 (Solar Mobil)	BMS concepts, battery/energy management, solar EV energy budgeting

[DESIGN] Member 1 owns Simulator + Track + Telemetry + Optimizer control loop; Member 2 owns Battery/Energy Model + Strategic Reserve + battery-model validation + real-world transfer's EV-domain grounding; the Energy Strategy Engine (Section 13) and Mathematical Formulation (Section 12) are joint-ownership.

4. What We Will and Will Not Claim

[DESIGN] Stated explicitly, up front, because it materially strengthens rather than weakens the pitch's credibility.

We will NOT claim: - Exact Haas VF-26 performance or physical behavior - Access to or use of proprietary Haas/Ferrari power-unit or battery data - A validated real F1 race strategy or a tool any real team would deploy as-is - FIA-certified simulation accuracy - A production-ready fleet-energy-optimization product

We WILL claim: - A research prototype that correctly implements the *publicly documented* 2026 FIA power-unit energy constraints (Section 2), with every non-public parameter explicitly labeled as an engineering assumption (Section 8) - A working, receding-horizon, uncertainty-aware optimizer that provably respects those constraints in simulation (Section 27's conservation/invariant tests) - A second, working configuration of the same optimization core applied to a real-world EV-fleet energy-budgeting scenario (Section 29), not merely a proposed application - Quantified, reproducible (Section 28) performance comparisons against a defined baseline hierarchy including an offline oracle upper bound (Section 18)

Part II — System Design

5. Product Concept and Two-Speed Simulation Architecture

Working name: TrackShift Energy Intelligence.

[DESIGN] Four functional layers: **World** (procedurally generated track/route + reduced-order vehicle/battery physics + scenario injector), **Estimation/Prediction** (state estimator + uncertainty-aware short-horizon forecaster, Section 14), **Decision** (receding-horizon MPC, Section 15), **Presentation** (dashboard, Section 32).

5.1 Two-speed simulation architecture

[DESIGN — correction from v1] A single simulator serving both “run 10,000 Monte Carlo races overnight” and “render a beautiful live demo” is the wrong architecture; those have opposed performance/fidelity tradeoffs. TrackShift Energy Intelligence therefore separates:

	Fast simulator (core)	**Visual simulator** (presentation layer)
Purpose	Training/calibration data generation, Monte Carlo sweeps (§23), optimizer tuning, ablations (§24), sensitivity analysis (§25)	Live demo, human-facing visualization, presentation mode
Implementation	Pure Python/NumPy numerical loop, no rendering	Same underlying physics/state, rendered via Canvas/WebGL in the dashboard (§32)
Throughput target	Thousands of full races in minutes-to-hours on a single machine	Real-time, one race at a time, judge-facing
Consumes	Same `vehicle_config` / `battery_config` / `regulation_config` as the visual simulator — **identical physics**, only the output/rendering path differs	Same

[DESIGN — explicit rejection] CARLA and Gazebo, despite being tools the team already knows well, are **not** used as the core simulator: they are robotics/perception-grade simulators built for a different problem (sensor-realistic 3D environments), and running thousands of energy-strategy races through either would be both unnecessary and far too slow. They remain available only as an optional, clearly-labeled Phase-5+ validation/visualization stretch (e.g. rendering one demo lap for visual polish), never as the system the optimizer or Monte Carlo pipeline actually runs against.

6. Haas-Inspired 2026 F1 Reference Vehicle Model

[DESIGN — naming correction from v1] This component is named the **“Haas-Inspired 2026 F1 Reference Vehicle Model”**, not “the Haas car” or a “Haas digital twin,” anywhere in the PRD, code, dashboard, or pitch materials. The PRD states explicitly, for the record:

> *"This model uses publicly available FIA/team-technical information where available (Section 2) and clearly labeled engineering assumptions for proprietary or unavailable parameters (Section 8). It is not intended to, and does not, reproduce the exact physical behavior of the real Haas VF-26."*

6.1 Haas-specific, publicly stated context

[FACT] Haas races under Toyota Gazoo Racing (TGR) Haas F1 Team branding for 2026, continues to run Ferrari power units through a technical partnership extending to 2028, fields drivers Esteban Ocon and Oliver Bearman, and its technical leadership (Team Principal Ayao Komatsu, Technical Director Andrea De Zordo) has publicly identified energy management as one of the season's defining engineering challenges. No Haas-specific numeric specification (mass distribution, battery chemistry/capacity beyond the shared regulatory Energy Store cap, drag coefficient, cornering-grip coefficients, exact deployment maps) is publicly released; these remain proprietary and are not fabricated here.

7. Known / Public-Estimate / Assumed / Derived Classification System

[DESIGN] Every vehicle and battery parameter in this project is classified into exactly one of four categories, consistently, everywhere it appears (dashboard, PRD, code comments, pitch deck):

Category	Definition
Official	Directly sourced from FIA/Haas/public regulation documentation, with a citable source (Section 2.2's provenance table is entirely this category)
Public estimate	Available from credible technical/journalistic sources but not an officially confirmed FIA/team figure (e.g. typical modern-F1 cornering-g or drag literature)
Engineering assumption	We choose a reasonable value ourselves because no public source exists (e.g. Haas-specific drag coefficient)
Derived	Calculated from other parameters already in one of the above three categories (e.g. corner-speed ceiling derived from assumed lateral-g and generated track radius)

8. Vehicle and Battery Parameter Table (Known/Assumed/Derived)

Parameter	Value used	Category	Basis
MGU-K deployment/regen power caps, Energy Store cap, recovery limits, deployment curve	Per Section 2.2	**Official**	FIA 2026 Technical Regulations (versioned, Section 2.1)
Minimum car weight	768–770 kg	**Official**	FIA 2026 Technical Regulations
Vehicle mass used in sim	800 kg	**Engineering assumption**, anchored to Official minimum	Regulatory minimum + reported industry early-season overweight margin
Drag coefficient × frontal area (CdA)	0.9–1.0 m^2 equivalent	**Public estimate**, scaled	Generic public F1 aero literature, adjusted by the regulation-stated ~drag-reduction target
Rolling resistance coefficient	0.012–0.015	**Public estimate**	Generic racing-slick tire literature, not Haas-specific
Peak lateral acceleration (cornering)	4.5–5.5 g	**Public estimate**	Publicly cited typical modern-F1 cornering-g literature

Parameter	Value used	Category	Basis
Peak longitudinal deceleration (braking)	~5–6 g	**Public estimate**	Publicly cited typical modern-F1 braking-g literature
Battery usable capacity	Modeled directly as the Official 4 MJ regulatory Energy Store cap	**Official** (no separate assumption needed)	Regulation is itself the constraint
Drive/regeneration round-trip efficiency	90% flat (MVP); 85–95% band, speed/temperature-dependent (Advanced)	**Engineering assumption**	Generic li-ion/motor-inverter efficiency literature
Thermal derate behavior	Not modeled in MVP; coarse proxy in Advanced model	**Engineering assumption**	Simplification decision, not FIA-specified
Corner-speed ceiling per track segment	`v = sqrt(μ_lateral · g · r)`	**Derived**	From assumed lateral-g (Public estimate) and generated segment radius (Design)
Per-lap achievable energy budget under baseline pace	Computed by the forecaster (§14)	**Derived**	From vehicle/battery model + track geometry

[DESIGN] This table, not narrative text, is the canonical reference; the full per-parameter appendix with sensitivity notes lives in Section 37 (Engineering Assumptions Appendix) so every number in the system is defensible to a judge asking “where did you get this?”

9. Track and Route Generation

[DESIGN — correction from v1] We do not use the term "random track" anywhere; the generator performs **constraint-based procedural circuit generation**. A track is a sequence of segments (straight, high-speed/medium/slow corner, hairpin, chicane, optional elevation change) connected by clothoid-like transition arcs so curvature is continuous — free-form random curvature sampling produces physically inconsistent, unrackable geometry and is explicitly rejected.

9.1 Generation constraints (all enforced, all reasons for resampling if violated)

- Closed circuit (start/end position and heading close up within tolerance)
- Continuous curvature (clothoid transitions, no discontinuous radius jumps)
- Minimum/maximum corner radius per corner class (Section 9.3)
- Realistic straight-length distribution (200–1,800 m, **Public estimate** from generic circuit-design literature)
- Bounded total track length (3.5–7.0 km, **Design** choice matching the broad range of real circuits)
- Bounded curvature-change rate (prevents "impossible" transitions no vehicle could follow)
- A feasible vehicle trajectory exists at all (validated by running the vehicle model over the generated track once at generation time and rejecting if the corner-speed/braking-zone geometry is infeasible under the vehicle's own dynamic limits — this is the "no impossible geometry" check, not a separate hand-written rule set)

9.2 Predefined track classes

[DESIGN — new in v2] Beyond raw procedural generation, the grammar supports six named classes (each a different distribution over segment-type frequency and corner-radius bands), so Monte Carlo and ablation results (Sections 23–24) can be reported *stratified by track character*, not just pooled:

Class	Character
High-speed	Long straights, high-radius corners, minimal braking-zone density
Technical	Short straights, dense mix of medium/slow corners, low top speed
Balanced	Even mix, representative "average" circuit
Heavy-braking	High frequency of hairpins/chicanes, large deceleration events
Energy-intensive	Long high-power straights, few regeneration opportunities — stresses the reserve logic
Energy-recovery-heavy	Frequent heavy-braking zones — favors aggressive deployment since regen opportunities are abundant

These classes are the primary axis for the ablation table in Section 24 and for reporting Monte Carlo results in Section 23.

9.3 Segment-type parameters

Segment type	Defining parameter	Typical range **[Public estimate, generic circuit literature]**
Straight	length	200–1,800 m
High-speed corner	radius	150–300 m
Medium corner	radius	60–150 m
Slow corner	radius	15–60 m
Hairpin	radius	8–15 m
Chicane	paired short-radius corners, opposite direction	2× 20–40 m radius, <50 m apart
Elevation segment (optional, Phase 3+)	gradient	±2–8%

9.4 Real-circuit validation (optional, Section 14 of prior review)

[DESIGN] Time-permitting, one or two openly-licensable real-circuit geometries (public track-map coordinate data, not proprietary telemetry) can be hand-encoded into the same segment representation to run as an external validation set — demonstrating "procedural tracks → generalization" and "real circuit → external validation" as two distinct evaluation claims (Section 22). This is explicitly P2 (Section 34), not required for the minimum winning prototype.

10. Vehicle Simulation Model

[DESIGN, unchanged in substance from v1, retained] Reduced-order point-mass longitudinal model with a corner-speed cap, not a full multi-body/tire-slip simulator — the correct fidelity level for an energy-strategy problem (full justification and rejected alternatives in the comparison table below).

- **Core state**: distance `s`, speed `v`, SOC, lap number, current deployment action
- **Longitudinal dynamics**: `F_drive = min(F_traction_limit, P_available(t)/v)`; drag `F_drag = 0.5·ρ·CdA·v²`; rolling resistance `F_roll = Crr·m·g`; net `a = (F_drive − F_drag − F_roll)/m`
- **Corner-speed limit**: `v_corner_max = sqrt(μ_lateral · g · r)`
- **Power/energy split**: ICE contribution bounded by the Official ~400 kW regulatory ceiling; MGU-K contribution bounded by the Official deployment curve (Section 2.1) **and** by currently available SOC — this split is exactly the optimizer's control variable (Section 16), never decided by the vehicle model itself

Candidate model	Verdict
Kinematic/geometric only	Rejected — cannot produce energy consumption
Point-mass longitudinal + corner-speed cap	**Adopted** — captures speed profile, power demand, braking/regen timing
Full nonlinear bicycle model w/ tire slip	Rejected for v1 — irrelevant fidelity for an energy-strategy problem; P2 stretch only if time allows, never load-bearing for the optimizer
Full multi-body (CARLA/Gazebo-grade)	Explicitly rejected — wrong tool, would consume the entire build (Section 5.1)

11. Battery / Energy Model

11.1 MVP model (deliberately minimal, per review guidance)

[DESIGN — scope correction] The MVP explicitly does **not** include detailed electrochemistry, full thermal modeling, cell-level simulation, or degradation — those are added only if evaluation evidence (Section 24's ablations) shows they materially change strategy quality. MVP state:

- `SOC_{t+1} = SOC_t − E_discharge(t) + η_regen · E_regen(t)`, clipped to `[0, ENERGY_STORE_CAP_MJ]`
- `E_discharge(t)` bounded by the Official deployment curve (Section 2.1) integrated over the step
- `E_regen(t)` bounded by the Official 350 kW regen cap and available braking energy
- Flat round-trip efficiency `η_regen = 0.90` (**Engineering assumption**)
- A configurable minimum reserve floor (ties to Section 15's Strategic Reserve)

11.2 Advanced model (Phase 3+, added only if justified by ablation evidence)

C-rate-dependent efficiency, a coarse thermal proxy (cumulative high-power-deployment heat state that temporarily de-rates max power past a threshold — **Engineering assumption**, not FIA-specified), and a simple cumulative-cycle degradation counter for narrative purposes only.

11.3 What the system computes every control step

Energy consumed (cumulative/per-lap), energy recovered, current SOC, predicted SOC trajectory to finish **with an explicit uncertainty band** (Section 14), remaining usable energy, dynamic safety reserve (Section 15), and strategically deployable energy.

11.4 Battery-model validation

[DESIGN — new in v2, addresses review point 26] *"How do we know our battery model isn't nonsense?"* is answered three ways, owned by the team's battery-domain member:

1. **Shape validation against public battery/EV literature** — the modeled SOC-depletion and regen-recovery curves are checked for qualitative consistency against publicly available battery discharge/charge curve shapes (not fit to any single proprietary dataset, since none is available for F1); this is a sanity check, not a calibration claim.
2. **Sensitivity analysis** (Section 25) — perturbing efficiency ±5%, drag ±10%, mass ±5%, regen ±10%, and checking the optimizer's strategy and outcome remain qualitatively stable, which is the actual claim of validity for a model built on Engineering Assumptions rather than proprietary data.
3. **Conservation/invariant automated tests** (Section 27) — energy can never be created; SOC always stays in `[0, ENERGY_STORE_CAP_MJ]`; regenerated energy per step never exceeds the physically available braking energy at that point on track.

12. Architecture Reframe: Not "AI → Decision"

[DESIGN — correction from v1] The control loop is reframed so optimization, not machine learning, is the load-bearing core, with ML confined to a clearly-scoped supporting role:

```
Telemetry
    ↓
State Estimation
    ↓
Prediction  (Physics-based model, optionally + ML — see §13)
    ↓
Constrained Optimization / MPC   ← the decision-making core
    ↓
Energy Deployment Decision  (with explicit uncertainty-aware margins, §14)
    ↓
Vehicle Simulation
    ↓
New Telemetry
    ↓
(repeat)
```

Machine learning, if used at all, sits **inside** the Prediction block only: `Physics + ML → Energy Prediction → MPC`. It is never positioned as "the AI that decides the strategy" — the optimizer decides; the predictor informs. This is both more technically defensible and a more accurate description of what the system actually does.

13. Prediction Approach — Comparison and Recommendation (Revised)

[DESIGN — explicit requirement, addresses review point 5] *The system must not use machine learning merely for the sake of using AI.* Every candidate approach is evaluated on its merits for this specific role (short-horizon energy/speed forecasting feeding the optimizer), not assumed:

Approach	Data needs	Explainability	Compute cost	Verdict for the Prediction block
Physics-based extrapolation (no learning)	None	Highest	Lowest	Strong default/fallback — always available, always explainable
Regression (linear/polynomial over recent telemetry features)	Small	High	Very low	**Primary candidate** — likely sufficient given the reduced-order vehicle model already constrains the achievable dynamics tightly

Approach	Data needs	Explainability	Compute cost	Verdict for the Prediction block
Gradient boosting (e.g. over engineered track/state features)	Moderate	Moderate	Low	Considered if regression underfits; still inspectable via feature importance
Gaussian process / other explicit uncertainty model	Moderate	Moderate–high	Moderate	Attractive specifically because it outputs a *native* predictive variance, directly feeding §14's uncertainty-aware reserve — a strong candidate if compute budget allows
Neural network (small GRU/LSTM)	Large(r)	Low	Moderate (train), low (infer)	Only justified if the simpler options are demonstrably insufficient (evaluated via the ablation table, §24) — not used by default
Dynamic Programming (as a full strategy, not just prediction)	None	High	High offline	Used as an **offline reference/near-oracle** (§18), not the online predictor
Reinforcement Learning (as a full strategy)	Very large	Low	High (train)	**Not used for live decision-making** (no native hard-constraint guarantee); optional Phase-5 comparison bar only

[DESIGN] Recommendation: start with physics-based extrapolation as the always-available fallback; escalate to regression, then (if the uncertainty-quantification value proves material) a Gaussian-process-style model, only as far as the ablation evidence (Section 24) justifies. **The MPC optimization core is fixed regardless of which predictor is used** — this is precisely what makes the architecture defensible: swapping the predictor never changes the guarantee that hard constraints are respected, because that guarantee lives in the optimizer, not the predictor.

14. Uncertainty-Aware Forecasting and Dashboard Representation

[DESIGN — new in v2, addresses review point 8] The system never treats a single-point forecast as ground truth. Every energy-requirement prediction is a **distribution**, minimally characterized by a mean and a standard deviation (or an explicit interval):

> Example: *Predicted remaining energy required to finish: 38.2 MJ. 95% interval: 36.8–39.7 MJ.*

This uncertainty (`σ_Ê(t)`) is not decorative — it is a direct input to the Strategic Reserve formulation (Section 15) and is shown explicitly on the dashboard's Strategy panel (Section 32), including a live "predicted vs. actual" comparison so a judge can see, quantitatively, when and how much the forecast missed:

- Mean absolute error (MAE) of the forecaster vs. realized telemetry
- Bias (systematic over/under-prediction)
- Error broken out by track class (Section 9.2) and by "clean race" vs. "race containing an injected event" — the second split is the important one, since a system's uncertainty-handling is judged specifically on how it behaves right after a disruption, not on average-case accuracy

15. Strategic Energy Reserve — Mathematical Formulation (Uncertainty-Aware)

At any race state, define `SOC(t)`, `Ê_required(t)` (point forecast of energy needed to finish under a baseline-competitive profile), and a **dynamic** safety reserve:

```
R_safety(t) = R_base + k · σ_Ê(t) + R_event_contingency(t)
```

where `R_base` is a small fixed floor covering residual model/measurement error, `σ_Ê(t)` is the forecaster's *current* predictive uncertainty (Section 14) — so the reserve automatically widens exactly when the forecaster is least confident, e.g. immediately after an unexpected event, before it re-converges — and `R_event_contingency(t)` is an optional extra margin the strategy manager can add ahead of a known energy-hungry upcoming feature.

```
Strategically Deployable Energy(t) = SOC(t) − Ê_required(t) − R_safety(t)
```

Worked example: `SOC = 62%`, `Ê_required = 43%`, `R_safety = 7%` → Deployable = `12%`. A safety-car-equivalent event lowers `Ê_required` to `37%` while `R_safety` stays `7%` (or briefly *widens* if `σ_Ê(t)` spikes right after the event is detected, then narrows again as the forecaster re-converges) → Deployable = `18%` (or momentarily less, if the reserve widened first) — this is the mechanism, not a lookup table.

16. Mathematical Formulation of the Optimization Problem

[DESIGN — new dedicated section, addresses review point 6, the single most important addition]

16.1 State

```
x(t) = [ SOC(t),
         v(t),                    # speed
         s(t),                    # position along track/route
         lap(t),
```

```
        distance_remaining(t),
        battery_temperature(t),  # Advanced model only (§11.2)
        track_state(t),          # grip/weather flag
        σ_Ê(t) ]                 # forecaster uncertainty
```

16.2 Control

```
u(t) = deployment power fraction ∈ [u_regen_max, u_deploy_max]
```

a single continuous decision variable per segment/time-step (Section 17), physically bounded at every instant by the Official deployment curve (Section 2.1) evaluated at `v(t)`.

16.3 Disturbances

```
d(t) = [ weather,                 # grip/rolling-resistance modifier
        traffic,                  # localized speed cap
        track_condition,          # e.g. limited-power-mode toggle, §2.4
        unexpected_energy_demand ]# step-change consumption event
```

Disturbances are not known in advance beyond the current forecast horizon; they are what the receding-horizon re-solve (Section 17) reacts to.

16.4 Constraints (all hard unless noted)

```
0 ≤ SOC(t) ≤ ENERGY_STORE_CAP_MJ                             for all t
u_deploy(t) ≤ DEPLOYMENT_CURVE(v(t))                 for all t   [regulatory, hard]
u_regen(t)  ≤ MAX_MGU_K_REGEN_POWER_KW               for all t   [regulatory, hard]
v(t) ≤ v_corner_max(s(t))                          for all t   [vehicle/track, hard]
Strategically Deployable Energy(t) ≥ 0                 for all t in horizon [soft-
then-hard, §15]
SOC(t_finish) ≥ R_base                                       [safety, hard]
```

The "Strategically Deployable Energy ≥ 0" constraint is treated as **soft** (a heavily-weighted penalty term) inside the rolling horizon during normal operation — allowing the optimizer to trade it off against lap-time objective at the margin — but becomes **hard** whenever the remaining-race feasibility check (Section 20) determines that violating it risks not finishing; this graduated treatment is itself part of the failure-mode design.

16.5 Objective

[DESIGN — explicit, addresses review point 7]

Primary objective (hard-constrained): minimize total race completion time, subject to all constraints in Section 16.4 being satisfied.

Objective function (per re-solve, over the rolling horizon `H`):

```
minimize   Σ_{t=0}^{H} [ lap_time_cost(v(t), u(t)) ]
          + λ_reserve · max(0, −Strategically_Deployable_Energy(t))   # soft penalty
        + λ_stability · |u(t) − u(t−1)|                         # discourage chattering
           + λ_uncertainty · σ_Ê(t)                                    # prefer lower-
uncertainty plans when tied on time
subject to constraints in §16.4
```

Priority hierarchy (addresses review point 7 directly — "optimal" is not ambiguous in this system):

1. **Hard constraints** (regulatory power limits, vehicle limits, SOC bounds, finish-with-reserve) — never violated in a feasible solve; if no feasible solution exists, the system reports infeasibility explicitly (Section 20) rather than silently violating a constraint.
2. **Primary soft objective** — minimize race completion time.
3. **Secondary soft objectives** — respect the reserve margin, prefer strategy stability (avoid needless chattering between deploy/conserve), prefer lower-uncertainty plans when two candidate plans are time-equivalent.

17. Receding-Horizon Control Loop and Action Representation

[DESIGN, refined from v1]

```
Current state x(t)
        ↓
Prediction engine forecasts d(t..t+H) and Ê(t..t+H) with uncertainty σ_Ê
        ↓
Solve the §16 optimization over horizon H (3–8 corners / ~500m–1.5km)
        ↓
Execute ONLY u(t), the first action of the optimized sequence
        ↓
Receive new telemetry → update x(t+1)
        ↓
Re-solve (repeat, every fixed distance/time step — tuned empirically in Phase 2)
```

Action space: `u(t)` is a **continuous** deployment-power fraction (never a hand-picked fixed schedule such as "use 20% battery every lap," which the theme brief explicitly warns against). For dashboard/narration purposes only (Section 32), the continuous output is post-hoc binned into five human-readable labels (Attack, Normal, Conserve, Coast, Harvest/Regen) — the optimizer itself never reasons in these bins.

18. Baseline Hierarchy (Including Oracle Upper Bound)

[DESIGN — expanded from v1, addresses review point 17] All baselines share the identical vehicle/battery model, track, and disturbance schedule — only the decision policy differs.

Level	Baseline	Policy
0	No optimization	Fixed maximum deployment until depleted, then forced near-zero — establishes the naive floor
1	Aggressive	`u = u_deploy_max` whenever SOC > 0, ignoring reserve
2	Conservative	`u` capped well below max at all times
3	Fixed heuristic	Predetermined per-lap schedule set once at race start from the average-case forecast, never updated

Level	Baseline	Policy
4	Physics-based optimization (no learned predictor, no uncertainty)	Full MPC/optimization core (§16) but with a physics-only point forecast and a fixed (not uncertainty-scaled) reserve — isolates the value of prediction quality and uncertainty-awareness (feeds directly into the §24 ablation table)
Proposed	Prediction + MPC/optimization, uncertainty-aware reserve	Full system as specified in §12–17
Oracle (upper bound, not a deployable baseline)	Offline optimization with full future knowledge of the entire race (all disturbances known in advance)	Establishes the theoretical best-case; the proposed system's result is reported as *"X% of oracle performance,"* which is a genuinely strong, judge-legible research metric because it honestly frames what "optimal" can mean for a system that, unlike the oracle, has no future information

19. Changing Race Conditions (Disturbance Model)

[DESIGN, retained from v1 with the §2.4 limited-power mode added as a regulation-grounded event]

Scenario	Mechanism modeled
Rain / wet track	Reduced `μ_lateral` (lower corner-speed ceiling), increased rolling resistance, reduced max deceleration
Safety car / reduced-pace period	Track-wide speed cap for a duration; lowers `Ê_required(t)`
Traffic / following car	Localized speed reduction over a segment
Unexpected consumption spike	Step-change added to per-segment energy demand
FIA “limited power” mode toggle (§2.4)	Switches `DEPLOYMENT_CURVE` to `LIMITED_POWER_MODE_CURVE` mid-race — a regulation-grounded, not invented, disturbance type
Battery/thermal derate (Advanced model only)	Temporary reduction of max deployment power

Core demo mechanic (unchanged principle, now more precisely instrumented): the pre-event MPC plan becomes measurably suboptimal → forecaster prediction error and `σ_Ê(t)` spike (visible on the Section 14 predicted-vs-actual panel) → the next scheduled re-solve produces a visibly different trajectory. No event-specific handling code is required; this is structural to the receding-horizon loop (Section 17).

20. Failure-Mode Handling and Infeasibility

[DESIGN — new in v2, addresses review point 29] The optimizer must never silently fail or silently violate a hard constraint. Every re-solve produces one of three outcomes:

1. **Feasible, normal** — a strategy exists that satisfies all hard constraints with the current reserve intact; execute `u(t)` as normal.
2. **Feasible, reserve-constrained** — a strategy exists but only by tightening toward maximum conservation; flagged on the dashboard as `CONSERVE — LOW ENERGY MARGIN` (Section 21) rather than presented identically to a normal decision.
3. **Infeasible** — no strategy satisfies the finish-with-reserve constraint under the current forecast (e.g. remaining achievable energy recovery cannot cover the remaining required energy even at maximum conservation). The system reports this explicitly:

```
Remaining energy: 12 MJ
Minimum required (max conservation): 15 MJ
STATUS: ⚠ Race completion infeasible under current forecast
Recommended action: Maximum conservation (best achievable, not a guaranteed finish)
```

This is materially more realistic than an optimizer that is implicitly assumed to always succeed, and it gives the dashboard (Section 32) a genuine “failure mode” view rather than only ever showing confident green numbers.

21. Explainability — the “Why” Layer

[DESIGN — new in v2, addresses review point 20, arguably the single highest-leverage judging feature] Every strategy decision the dashboard shows must be traceable, in the UI itself, to the specific numbers that produced it — never a bare label with no justification:

```
DEPLOY
SOC: 68%              Projected required energy: 54%
Reserve: 7%            Available strategic energy: 7%
Opportunity: high-performance straight ahead
Expected time gain: 0.31 s
```

```
CONSERVE — LOW ENERGY MARGIN
SOC: 38%               Projected required energy: 34%
Reserve: 7%             Projected deficit: 3%
```

22. Strategy Confidence (Not a Generic "AI Confidence" Score)

[DESIGN — new in v2, addresses review point 21] Rather than an opaque "AI confidence: 94%," the dashboard computes and shows a concrete, physically meaningful robustness readout:

```
Strategy robustness: HIGH
Predicted finish SOC: 9.1%      Uncertainty: ±1.4%
Worst-case finish SOC: 7.7%     Required reserve: 6.0%
→ SAFE TO DEPLOY
```

versus, when the margin is thin:

```
Strategy robustness: LOW
Worst-case finish SOC: 4.1%     Required reserve: 6.0%
→ CONSERVE — LOW ENERGY MARGIN
```

This number is derived directly from `σ_Ê(t)` and the Section 15 reserve formulation — it is never a fabricated or cosmetic "AI" score.

23. Strategy-Event Timeline

[DESIGN — new in v2] The dashboard logs a scrollable, judge-replayable timeline of every strategy-relevant event and decision change, e.g.:

```
LAP 3   CONSERVE
LAP 5   DEPLOY
LAP 7   REGEN
LAP 9   RAIN DETECTED
LAP 9   REPLANNING
LAP 10  CONSERVE
LAP 13  ATTACK
```

This both supports live narration during the demo (Section 33) and doubles as a post-race audit log for the evaluation pipeline (Section 21's Metrics).

Part III — Evaluation and Research Methodology

24. Metrics

[DESIGN — expanded from v1, addresses review points 18–19] Constraint violations and computational performance are first-class metrics, not afterthoughts — a strategy that is faster but fails to finish legally is not a better strategy.

24.1 Feasibility metrics (checked first, before any performance comparison is meaningful)

Metric	Definition
Race completion rate	Fraction of runs that finish with `SOC(t_finish) ≥ 0`
SOC constraint violations	Count of steps where `SOC(t)` left `[0, ENERGY_STORE_CAP_MJ]`
Regulatory power violations	Count of steps where `u(t)` exceeded the Official deployment curve (Section 2.1)
Reserve violations	Count of steps where Strategically Deployable Energy `< 0` while the hard-constraint mode (§16.4) was active
Optimization failures / infeasible solves	Count of re-solves returning the Section 20 infeasible outcome
Replanning failures	Cases where a re-solve failed to converge within the compute-time budget

24.2 Performance metrics (only compared across runs that passed §24.1)

Total race time; average lap time; energy consumed/remaining at finish; SOC trajectory shape (qualitative); strategy stability (action-to-action variance, excluding legitimate event-driven changes); performance delta vs. each baseline in the Section 18 hierarchy, **including percent of Oracle performance**; robustness across held-out/unseen tracks (Section 25).

24.3 Computational performance

[DESIGN — new in v2, addresses review point 19] Measured, never asserted without measurement:

- Prediction (forecaster) latency per call
- Optimization (MPC solve) latency per re-solve
- Total decision latency (telemetry → executed action)
- Fast-simulator throughput (races/minute) for Monte Carlo purposes

Only after these are actually measured does the PRD/pitch make a claim such as *"strategy decision computed in <X ms"* — no number is asserted here in advance of measurement.

24.4 Prediction-quality metrics (feeds §14's dashboard panel)

Mean absolute error and bias of the forecaster vs. realized telemetry, broken out by track class (Section 9.2) and by clean-race vs. event-containing race.

25. Train / Validation / Unseen-Test Separation

[DESIGN — corrected from v1, addresses review point 11] Because all data is self-generated, this separation must be a hard architectural rule, not an afterthought:

```
TRAIN        – generator seeds 1 – 7,000
VALIDATION   – generator seeds 7,001 – 8,500
UNSEEN TEST  – generator seeds 8,501 – 10,000
```

Training seeds are used for any learned-predictor fitting and iterative tuning; validation seeds are used only for hyperparameter/threshold tuning (horizon length, reserve constants, λ weights in Section 16.5); the unseen-test range is touched only for final reported numbers and, ideally, a subset of live demo tracks. **[DESIGN — strengthened]** Where feasible, the unseen-test set should also include at least one *track class* (Section 9.2) held out entirely from training/validation, not merely held-out seeds within an already-seen class distribution — a stronger generalization claim than seed-holdout alone.

26. Monte Carlo / Large-Scale Simulation Methodology

[DESIGN, retained with clarified reporting requirements] The fast simulator (Section 5.1) sweeps track class/seed, starting SOC, race length, weather scenario, and injected-event schedule, running all baseline-hierarchy policies (Section 18) under identical conditions per configuration.

Reporting requirement: aggregate statistics (mean, variance, worst-case) are reported **stratified by scenario type** — “clean race” vs. “race containing ≥1 injected event” — because the adaptive system’s value proposition is specifically its advantage under disruption; a pooled average across mostly-clean races would understate it. Target scale: hundreds to low thousands of runs, computed entirely pre-event (Section 35), never attempted live during the 24-hour sprint beyond a small confirmatory subset.

27. Ablation Studies

[DESIGN — new in v2, addresses review point 12] Each component’s marginal contribution is isolated by successive addition, run across the full Monte Carlo scenario bank (Section 26) and reported in one table:

System variant	Race time (vs. Oracle %)	Constraint violation rate	Robustness (unseen-track performance delta)
Fixed heuristic (Baseline 3)	—	—	—
Physics-only optimizer, no prediction, fixed reserve (Baseline 4 minus uncertainty)	—	—	—
+ Learned/statistical prediction (still fixed reserve)	—	—	—
+ Uncertainty-aware dynamic reserve (§15)	—	—	—

System variant	Race time (vs. Oracle %)	Constraint violation rate	Robustness (unseen-track performance delta)
Full proposed system (prediction + MPC + uncertainty + adaptive replanning)	—	—	—

Numeric cells are populated from actual Phase 2 experiments (Section 26), never estimated — this table's purpose is to let the team (and judges) see precisely which design decision earned its keep, directly supporting Section 4's "we will not claim more than we measured" posture.

28. Sensitivity Analysis

[DESIGN — new in v2, addresses review point 28] Because several vehicle/battery parameters are Engineering Assumptions (Section 8), not Official values, robustness to being *wrong* about them is itself evaluated: perturb battery efficiency ±5%, drag ±10%, vehicle mass ±5%, and regeneration efficiency ±10%, then re-run the Monte Carlo bank and confirm the proposed system's *qualitative* strategy behavior and relative advantage over baselines remain stable. This produces a genuine "robustness to model uncertainty" claim, which is materially stronger evidence than defending each assumed number individually.

29. Conservation and Invariant Tests (Automated)

[DESIGN — new in v2, addresses review point 27] Because the simulator is physics-based, these are implemented as automated unit/integration tests run on every change, not manual spot-checks:

- **Energy conservation** — cumulative `E_discharge − η_regen·E_regen` must equal the SOC delta at every step; energy cannot appear from nowhere
- **SOC bounds** — `0 ≤ SOC(t) ≤ ENERGY_STORE_CAP_MJ` at every step, always
- **Power bounds** — `u(t)` never exceeds the Official deployment curve (Section 2.1) at the corresponding speed
- **Regeneration physical limit** — `E_regen(t)` never exceeds the energy physically available from braking at that point on track (i.e. never regenerates more than the kinetic energy actually being shed)
- **Race-completion honesty** — the vehicle cannot be reported as having finished with less than the required reserve energy unless the scenario is *intentionally* simulating a failure/infeasible case (Section 20)

30. Reproducibility and Experiment Tracking

[DESIGN — new in v2, addresses review points 38–39] Every experiment run stores a complete, reproducible record:

```
random_seed
track_parameters (class, seed, generated geometry hash)
vehicle_parameters (config version)
battery_parameters (config version)
weather / disturbance schedule
initial_SOC
race_length
optimizer_configuration (horizon length, λ weights, solver version)
```

```
regulation_config_version  (§2.1's FIA_REGULATION_VERSION)
model_version
```

Each run is assigned a unique experiment ID for reference in the Monte Carlo results, ablation table, and pitch materials, e.g.:

```
EXP-000182
Track: TECHNICAL-47 (class: Technical, seed: 47)
Vehicle: HAAS-REF-01
Race: 20 laps, SOC 100%, weather DRY
Optimizer: MPC-v0.4
Seed: 918273
```

This makes every reported number in the pitch traceable back to an exact, re-runnable configuration — directly supporting Section 4's credibility posture and pre-empting "can you reproduce that number" questions from judges.

Part IV — Real-World Transfer

31. Transfer Architecture: One Optimization Core, Two Working Configurations

[DESIGN — corrected from v1, addresses review point 22] The prior draft's framing ("this could be used for EV fleets") is replaced with a concrete architectural commitment: the **same optimization core**, unmodified, is instantiated against two configurations, and both are run to demonstrate the transfer live, not asserted:

```
              OPTIMIZATION CORE  (§16 mathematical formulation, unmodified)
                           │
              ┌────────────┴───────────┐
              ▼                        ▼
         F1 CONFIG                EV FLEET CONFIG
    ─────────────────        ──────────────────────
    Race energy budget        Fleet/vehicle energy budget
    Finish-line deadline      Delivery-window deadline
    Track (generated)         Route (generated, same segment grammar)
    Vehicle: F1 reference     Vehicle: delivery EV reference
    Regen: braking/MGU-K      Regen: minor (downhill/braking only)
    Disturbance: rain/SC      Disturbance: traffic/detour/extra stop
```

Only the **configuration** changes (vehicle/battery parameters, route generator variant, disturbance types); the state representation (Section 16.1), control representation (Section 16.2), constraint structure (Section 16.4), and solver (Section 16) are identical. This is the strongest possible evidence, to a judge, that "the principle transfers" — it is a second working system, not a slide.

[DESIGN] Route generation reuses the Section 9 segment grammar directly: a delivery route is a sequence of road-link segments (distance + speed limit, replacing corner-radius-derived speed limits) with the delivery deadline playing the role of the finish line, and an unplanned extra stop or road closure playing the role of the safety-car/rain scenario injector (Section 19).

32. Why This Domain: Candidate Evaluation

Domain	Real-world impact	Data availability	Feasibility (24h-adjacent)	India relevance	Verdict
EV delivery-fleet battery-budget management	High — actively named, funded pain point in Indian quick-commerce/logistics right now	Good — synthetic routes via the reused generator; public commentary on delivery patterns	High — same state/control/constraint structure already built	**Very high**	**Primary, built**

Domain	Real-world impact	Data availability	Feasibility (24h-adjacent)	India relevance	Verdict
Microgrid/renewable-storage dispatch	High but diffuse	Moderate	Moderate — multi-period LP/MPC, weaker fit to the single-vehicle receding-horizon structure already built	Moderate	Secondary framing only, not built
EV charging/battery-swap-station scheduling	Moderate-high	Moderate	Moderate — closer to queuing than SOC-trajectory	High	Not built
Data-center energy management	Moderate	Low (proprietary)	Low	Low	Rejected

33. India-Specific Concrete Scenario

[DESIGN — corrected from v1, addresses review point 23: a concrete scenario, not a generic "EVs are growing" claim]

> *A last-mile delivery fleet operator (electric two/three-wheelers, quick-commerce or e-commerce delivery) must complete a multi-stop route within a delivery window, with limited depot charging capacity and no guarantee of mid-route charging access. The operator needs each vehicle to depart with sufficient charge to complete its assigned route without stranding, while minimizing unnecessary conservative over-charging (which wastes depot charging-slot time and peak electricity demand) and unnecessary aggressive energy use (which risks a mid-route failure to deliver).*

[FACT] This is not a hypothetical: Indian last-mile/quick-commerce operators are actively building "energy intelligence" tooling specifically to predict battery performance, optimize charging behavior, and reduce range anxiety, because compressed delivery timelines (now down to minutes in quick-commerce) make charging downtime a direct revenue cost; industry and fleet-management sources explicitly recommend incorporating dynamic SOC, EV-specific routing parameters, and range-aware dynamic rerouting into fleet software — a close structural match to this project's reserve-and-replan formulation. Electric two/three-wheeler delivery fleets (Zomato/Swiggy/Zypp-style operators) are the concrete, high-volume Indian EV segment this targets.

Who would use it: fleet operations/dispatch teams at last-mile logistics or quick-commerce operators. **What resource constraint exists:** limited depot charging slots/power, battery capacity, and delivery-window deadlines, simultaneously. **What measurable benefit occurs:** fewer mid-route failures-to-deliver (stranded vehicles), less wasted conservative charging time, and reduced peak-demand charging bunching at the depot (Section 34).

34. Impact Calculation Framework

[DESIGN — new in v2, addresses review point 24; numbers below are placeholders showing the required format, populated only from actual Section 26-style experiments run on the EV-fleet config, never asserted without measurement]

Motorsport config:

```
Compared with fixed-heuristic baseline:
Race time            ↓ X%
Energy violations     ↓ Y%
Strategy/infeasibility events   ↓ Z%
```

EV-fleet config:

```
Compared with a fixed/uniform charging & deployment strategy:
Vehicles meeting departure-SOC requirement   ↑ X%
Mid-route stranding incidents                ↓ Y%
Depot peak-charging-demand bunching          ↓ Z%
```

Both tables are populated identically from the reproducible experiment framework in Section 30 — an `EXP-####` ID exists for every number quoted here, exactly as for the motorsport results.

Part V — Dashboard and Demonstration

35. Dashboard Design Philosophy

[DESIGN — correction from v1, addresses review point 36] The dashboard reads as a **race-engineering terminal**, not an "AI product" — this is a deliberate, stated design constraint:

Avoid: glowing/futuristic effects, decorative "AI POWER" branding, generic animated confidence meters, gauges added for visual density rather than information value. **Prioritize:** track/route view, live telemetry, the Section 21 "why" explanation layer, energy graphs, prediction-vs-actual (Section 14), and the baseline comparison — every element traceable to a specific equation or constraint in Section 16, per the closing principle of the original PRD's Section 25, retained here unchanged.

36. Dashboard Panels

1. **Race/route state header** — lap or delivery-stop progress, position, speed, SOC, battery power, cumulative energy consumed/recovered.
2. **Strategy panel** — current decision with the Section 21 "why" breakdown, Section 22 robustness readout, `σ_Ê(t)` uncertainty indicator, estimated finish time/SOC.
3. **Track/route visualization** — vehicle or delivery-vehicle position, colored by the current deployment band (Attack/Normal/Conserve/Coast/Harvest — Section 17).
4. **Energy graphs** — SOC vs. distance/time, power vs. time, **predicted-vs-actual energy** (the panel that visually proves a replan happened, Section 14), strategy-change markers aligned to the Section 23 timeline.
5. **Performance comparison panel** — proposed system vs. the full Section 18 baseline hierarchy (including Oracle), for both the live scenario and the pre-computed Monte Carlo aggregate (Section 26).
6. **Failure-mode indicator** (Section 20) — visible, not hidden, whenever the system is in a reserve-constrained or infeasible state.

37. Final Demonstration Script

[DESIGN — reordered from v1 per review point 31: start with the problem, not the dashboard]

1. **Set the race** — 20 laps, 100% starting energy, all four/five policies (Section 18) configured identically.
2. **Show the baseline first** — the fixed-heuristic strategy running alone, so the audience has a reference before seeing the proposed system.
3. **Run** — car begins consuming energy; live telemetry visible.
4. **Introduce an unexpected condition** — rain, safety car, or the FIA limited-power-mode toggle (Section 19).
5. **Show the original plan breaking** — predicted-vs-actual panel diverges; projected finish becomes infeasible or reserve-constrained (Section 20).
6. **System detects it** — `σ_Ê(t)` spike visible, dashboard flags `REPLANNING`.
7. **New strategy deployed** — conserve → deploy → conserve trajectory visibly different from the pre-event plan; explain via the Section 21 "why" panel.
8. **Finish** — final state, feasibility outcome, energy remaining.
9. **Compare** — full baseline hierarchy including Oracle percent (Section 18/24), and the pre-computed Monte Carlo aggregate (Section 26) so this isn't a cherry-picked single run.
10. **Transfer** — switch the same dashboard/optimizer to the EV-fleet configuration (Section 31) and run an equivalent disruption-and-replan demo live.

38. Demo Modes

[DESIGN — new in v2, addresses review point 32: eliminate live-generation risk during judging]

Mode	Purpose
Demo Scenario A	Deterministic, rehearsed, “perfect showcase” scenario — the primary judged run
Demo Scenario B	Deterministic failure/replanning scenario — specifically built to trigger Section 20’s infeasible/reserve-constrained state at least once
Demo Scenario C	Real-world transfer run (Section 31)
Live Random Scenario	Available on request for judges who want to see true generalization on a track/seed not in the rehearsed set — this is the credibility backstop for the “not overfit” claim (Section 25)

Demo Scenarios A–C are frozen and rehearsed well before the event (Section 39); the Live Random option carries acceptable risk precisely because it is optional, not the primary judged path.

Part VI — Execution

39. Priorities: P0 / P1 / P2

[DESIGN — new in v2, addresses review points 34–35] Every feature in this PRD is tagged so the team never spends scarce hours on P2 polish while a P0 item is unfinished.

P0 — must have (core judging value): Haas-inspired reference vehicle (Section 6) with the Section 2 regulation config; one procedural track (Section 9); MVP battery model (Section 11.1) with conservation tests (Section 29); fast simulator (Section 5.1); Baseline hierarchy levels 0–3 (Section 18); MPC optimizer against the Section 16 formulation; live SOC + strategy visualization; at least one baseline-vs-proposed comparison with real numbers.

P1 — should have (strong enhancement): Uncertainty-aware prediction and dynamic reserve (Sections 14–15); scenario injector and live replanning demo (Section 19); full dashboard panels (Section 36); Monte Carlo aggregate results (Section 26); ablation table (Section 27); Oracle baseline (Section 18); EV-fleet transfer config (Section 31) run live.

P2 — nice to have (only if time remains): Advanced battery model terms (Section 11.2); real-circuit validation (Section 9.4); RL comparison bar (Section 13); sensitivity analysis full sweep (Section 28) beyond a spot check; CARLA/Gazebo visualization polish; microgrid secondary-framing slide (Section 32).

40. Minimum Winning Prototype (Hard Fallback)

[DESIGN — new in v2, addresses review point 34] If everything else slips, this exact set must still exist and be demoable:

> Haas-inspired reference vehicle **+** one procedural track **+** MVP battery model **+** race simulation **+** fixed-heuristic baseline **+** MPC optimizer **+** live SOC display **+** strategy visualization **+** one measurable baseline-vs-proposed comparison.

Everything else in this PRD is P1/P2 relative to this floor.

41. Phased Roadmap

[DESIGN — reaffirms and tightens v1's phasing per review point 33: "do not plan to develop the core optimizer during the 24 hours"]

Phase 0 — Research/Foundation (pre-event). Finalize `vehicle_config` / `battery_config` / `regulation_config` (Sections 2, 8) with FACT/ASSUMPTION/DERIVED labels; repo, schema, and vehicle+battery model skeleton with the Section 29 conservation tests wired in from day one.

Phase 1 — MVP (pre-event). Everything in Section 40's minimum winning prototype, fully working end-to-end, including the fast simulator (Section 5.1).

Phase 2 — Intelligent Strategy (pre-event). Prediction block (Section 13, starting with physics-based/regression); full MPC core against the Section 16 formulation; Baseline hierarchy levels 4 and Oracle (Section 18); train/val/test seed split (Section 25); Monte Carlo runner executed at scale now, offline (Section 26) — the bulk of the numbers shown live at the event should already exist by the end of this phase.

Phase 3 — Adaptive Intelligence (pre-event, ideally; first to slip if behind). Scenario injector (Section 19) and uncertainty-aware reserve (Sections 14–15) confirmed across multiple seeds, not one lucky run; Advanced battery-model terms only if ablation evidence (Section 27) justifies them.

Phase 4 — Final Product Polish (pre-event where possible; primary focus of sprint hours 1–14). Full dashboard (Section 36); EV-fleet transfer config (Section 31); Demo Scenarios A–C (Section 38) rehearsed.

Phase 5 — 24-Hour Sprint.

Freeze before the 24 hours	Build during the 24 hours	Explicitly avoid on-site
Vehicle/battery models, all configs	Dashboard polish and EV-fleet re-skin	Building a new physics fidelity level from scratch
Track/route generator	Live demo scenario scripting/rehearsal	Training a larger/new ML predictor under time pressure
MPC core, tuned horizon	Backend/frontend integration, WebSocket wiring	The RL comparison stretch, unless hours ahead of schedule
Scenario injector	Bug-fixing integration issues	Sourcing any new external dataset not already vetted (Section 32)
Monte Carlo + ablation results (the actual numbers)	Presentation deck, live-demo runbook	Re-running the full Monte Carlo sweep live (use precomputed results; small confirmatory subset only)
Baseline + Oracle implementations		Changing the optimizer's constraint formulation the night before presenting

42. Risk Register

Risk	Probability	Impact	Mitigation	Contingency
Lack of real Haas vehicle data	Certain	Low if handled transparently	Section 6/8's public-regulation-grounded, explicitly-labeled model	Present the K/A/D labeling itself as evidence of rigor if challenged
FIA regulation figures change again before the event	Medium (regulations have already changed twice in 2026)	Medium	Config-driven, versioned `FIA_REGULATION_VERSION` (Section 2.1), never hard-coded	Re-verify and update the config in the days before the event; the architecture already isolates this to one file
Simulation judged "unrealistic"	Medium	Medium	Cited methodology throughout Sections 9–10; K/A/D classification visible and inspectable	Keep config files open during Q&A

Risk	Probability	Impact	Mitigation	Contingency
Overfitting to generated tracks	Medium	Medium	Seed-range + track-class holdout (Section 25)	Live-generate an unseen seed during judging as direct rebuttal
Optimization latency too high for real-time demo	Medium	Medium	Measured (not assumed) latency budget, horizon tuned in Phase 2 (Section 24.3)	Rely on precomputed Monte Carlo/ablation results; only one scenario needs to solve live
Battery-model inaccuracies	Medium	Low–Medium	MVP intentionally simple (Section 11.1) + validation (Section 11.4) + sensitivity analysis (Section 28)	Present MVP/Advanced distinction proactively
Overengineering (Advanced-model terms consuming schedule)	Medium	Medium	Explicit P0/P1/P2 tagging (Section 39); Advanced terms gated behind ablation evidence	Cut per the Phase 5 freeze list (Section 41)
Dashboard outweighing the core algorithm	Medium	Medium	Architectural separation (dashboard is read-only w.r.t. the control loop); "why" panel narrates the math, not just the UI (Section 21)	Cut dashboard polish before optimizer correctness if forced to choose
24-hour implementation risk generally	High	High	Explicit freeze/build/avoid table (Section 41); Minimum Winning Prototype floor (Section 40)	Maintain a demoable checkpoint at the end of every sprint block

Risk	Probability	Impact	Mitigation	Contingency
Real-world transfer appearing superficial	Medium	High (explicitly weighted in TrackShift's judging philosophy)	Shared optimization core with two working configs (Section 31), not a slide	If the EV-fleet run can't be fully live, present the shared-core config diff plus a pre-run recorded clip as evidence of working feasibility
Difficulty proving measurable impact	Medium	Medium–High	Reproducible experiment framework (Section 30) + ablation table (Section 27) computed ahead of time	Lead the pitch with aggregate/ablation numbers, not only the single live-demo anecdote
Optimizer reports infeasible mid-demo unexpectedly	Low–Medium	Medium (could look like a bug rather than a feature)	Section 20's infeasibility handling is a *designed* feature, not an error state — Demo Scenario B (Section 38) is specifically built to show it on purpose	Narrate infeasibility, when it occurs, as the system correctly refusing to overpromise — consistent with Section 4's honesty framing

43. Engineering Assumptions Appendix (Consolidated Reference)

[DESIGN — new in v2, addresses review point 37] Every Engineering Assumption or Public Estimate in this PRD is additionally logged in this single appendix table for quick judge-facing lookup, in the following format (populated from Sections 2, 8, and 9 — shown here as the required schema, cross-referencing the parameter tables already defined rather than duplicating them):

```
ASSUMPTION-001
Parameter: Vehicle mass used in sim
Value: 800 kg
Category: Engineering assumption (anchored to Official minimum weight)
Reason: Regulatory minimum (768–770 kg, Official) + reported early-season
        overweight margin
Source: §8 parameter table
Sensitivity: Tested ±5% per §28
Status: Assumption
```

```
ASSUMPTION-002
Parameter: Drag coefficient × frontal area (CdA)
Value: 0.9–1.0 m² equivalent
Category: Public estimate, scaled
```

```
Reason: Generic F1 aero literature adjusted by regulation-stated drag reduction
Source: §8 parameter table
Sensitivity: Tested ±10% per §28
Status: Public estimate
```

(…continues for every row of Section 8's parameter table; maintained as a living appendix, not re-derived ad hoc during the event.)

44. References and Source Notes

Categories of public source drawn on throughout this PRD, current as of August 2026:

- **TrackShift / Plaksha University / Mphasis Foundation** — official program materials and press coverage of the TrackShift Innovation Challenge and its partnership with the Mphasis F1 Foundation and Toyota Gazoo Racing Haas F1 Team.
- **FIA 2026 Formula 1 Technical Regulations, including the August 5, 2026 amendment** — Formula1.com regulation explainers and terminology updates; Honda Racing Corporation's public regulations overview; Red Bull Racing and F1 Las Vegas Grand Prix "guide to the 2026 regulations"; ESPN, The Race, Motorsport.com, GPFans, Speedcafe, Raceteq, Motorsport Technology, and PaddockNews24 reporting on MGU-K power/energy limits, minimum weight, dimensions, and the mid-season and August 2026 rule refinements (Article C5.2.8 iii, Article C5.2.10 ii, Article B7.2). PlanetF1 coverage of the 2027 power-split rule change, explicitly noted as **out of scope** for this 2026-dated PRD.
- **Haas F1 Team / Toyota Gazoo Racing Haas F1 Team** — Formula1.com, ESPN Africa, and flashscore.com coverage of the Haas VF-26 launch, 2026 driver lineup, Ferrari power-unit partnership, and public statements from Team Principal Ayao Komatsu and Technical Director Andrea De Zordo.
- **Energy-management control literature** — published research on MPC-based energy management for hybrid/electric vehicles (including learned-prediction-fused MPC and scenario/data-based MPC), and MPC-based race-strategy optimization for electric endurance racing accounting for competitor interaction and pit-stop timing (arXiv, MDPI, ScienceDirect).
- **Real-world transfer domain grounding** — public reporting on India's last-mile/quick-commerce EV delivery ecosystem, battery/energy-intelligence tooling, range anxiety, and EV-fleet telematics/routing practices, plus a 2025 arXiv survey on data-driven optimization of EV charging infrastructure, scheduling, and fleet management.

Regulation-currency caveat (repeated from Section 2.5): FIA 2026 technical regulations have been amended multiple times within the season itself, most recently August 5, 2026. `FIA_REGULATION_VERSION` (Section 2.1) must be re-verified against the latest FIA/Formula1.com publication in the days immediately before the event.